EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM

To create a 3Dmodel from the given isometric drawing using fusion 360 tool.

REQUIREMENTS

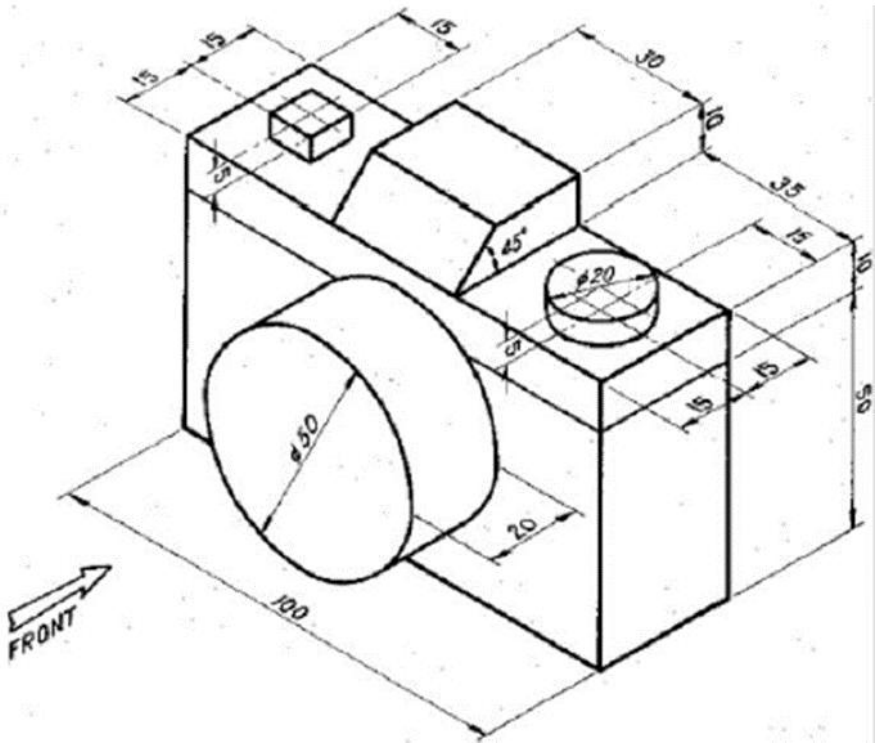
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

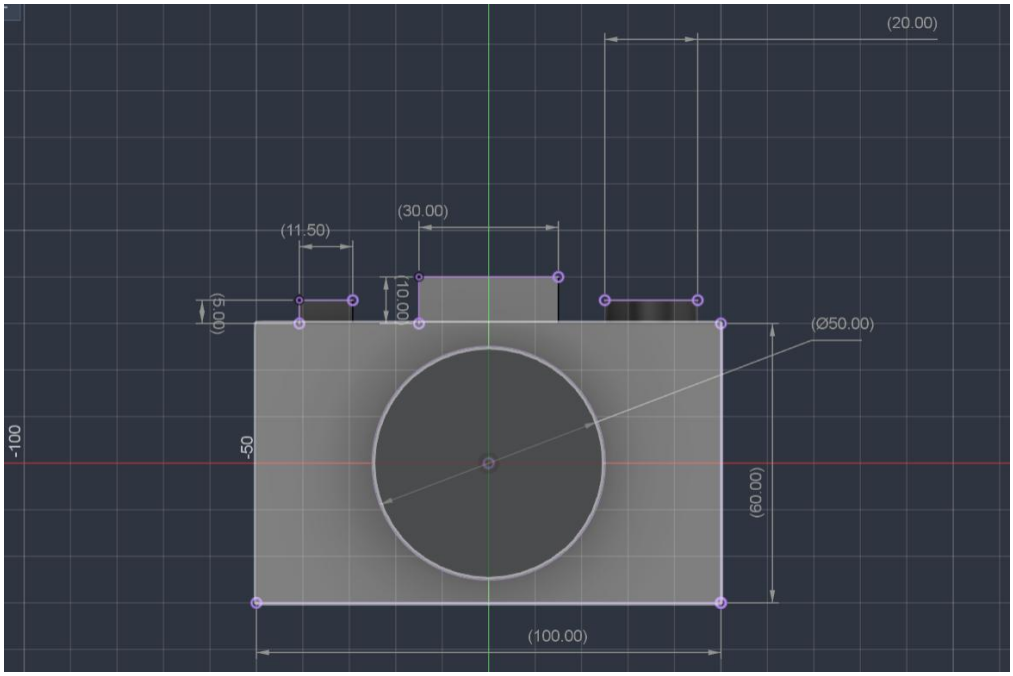
PROCEDURE

1. **Start a New Design & Base Sketch:** Open Fusion 360, create a new design, select **Create Sketch**, and choose the Front (XY) Plane.
2. **Draw the Main Body Profile:** Select the **Centre Rectangle** or **2-Point Rectangle** tool and sketch a rectangle measuring **100 mm wide** by **50 mm high**.
3. **Extrude Main Body:** Finish the sketch and use the **Extrude (E)** tool to extrude the rectangle **35 mm** backward to create the main camera block.
4. **Sketch the Front Lens:** Create a new sketch on the front face of the extruded block. Draw a circle with a diameter of **50 mm** centred horizontally.
5. **Extrude the Lens:** Finish the sketch and use **Extrude (E)** to join a cylinder **20 mm** out from the front face.
6. **Sketch Top Flash/Viewfinder Profile:** Select the top face of the main body and start a new sketch. Sketch a trapezoidal cross-section **30 mm wide** with **45° angled sides**.
7. **Extrude the Viewfinder Hump:** Extrude the trapezoid profile upward by **10 mm** across the top depth of the camera.
8. **Sketch the Left Button:** Create a sketch on the top-left surface. Draw a **15 mm × 15 mm square** located **15 mm** from the outer edges.
9. **Extrude the Left Button:** Use **Extrude (E)** to pull the square profile upward by **5 mm**.

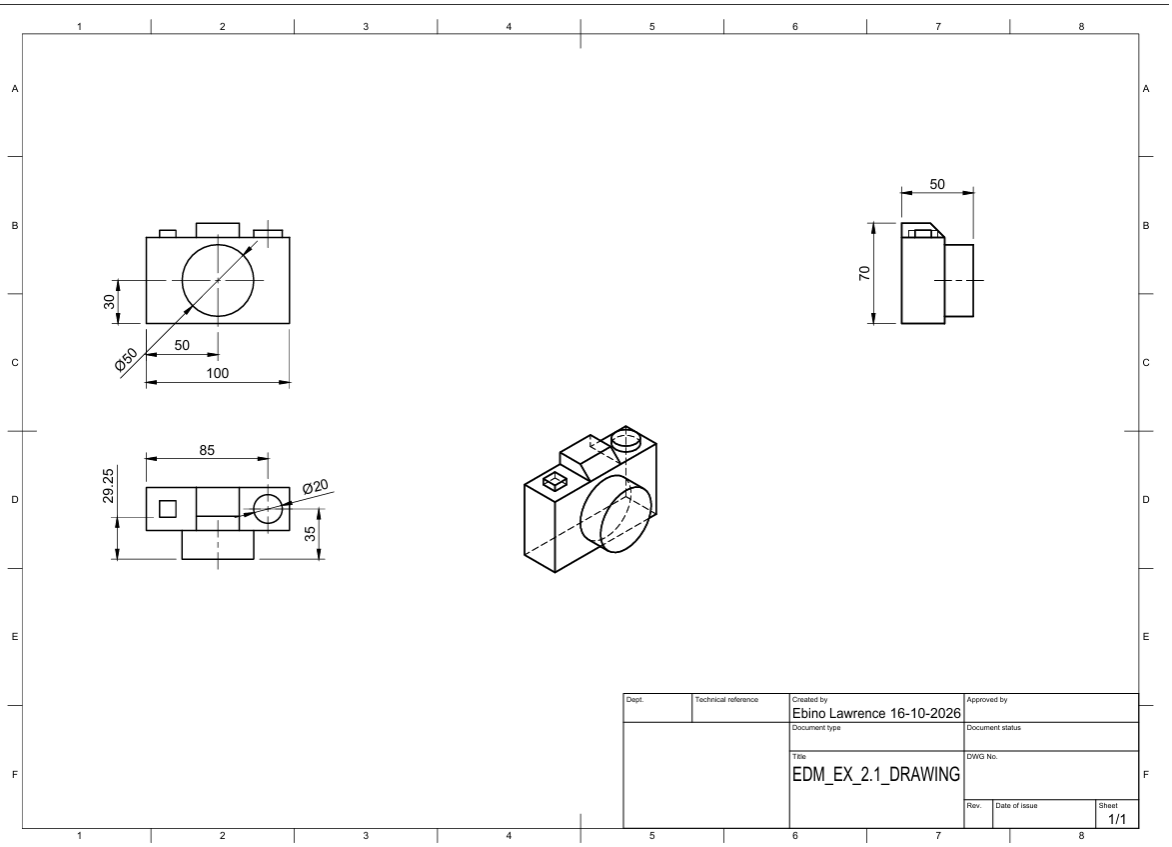
10. **Sketch & Extrude Top Right Dial:** Create a sketch on the top-right surface, draw a **Ø20 mm circle** spaced **15 mm** from the edges, and extrude it upward by **10 mm**.

INPUT





OUTPUT



RESULT

Thus, the 3D model is created from given isometric drawing.